CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The four spheres of the Earth system are the geosphere, hydrosphere, atmosphere and: 1

(a) stratosphere **(b)** troposphere **(c)** lithosphere only **(d)** biosphere

2. The surface that reflects the most solar radiation is: 1

(a) forest **(b)** dark soil **(c)** fresh snow **(d)** ocean

3. The ozone layer is present in the: 1

(a) stratosphere **(b)** troposphere **(c)** geosphere **(d)** mesosphere

4. The continuous movement of water between the oceans, atmosphere and land is the: 1

(a) water (hydrological) cycle **(b)** rock cycle
(c) carbon cycle **(d)** nitrogen cycle

5. The solid Earth including rocks, soil and the interior is the: 1

(a) geosphere **(b)** hydrosphere **(c)** atmosphere **(d)** biosphere

6. The surface that reflects the least solar radiation is: 1

(a) ocean water or forests **(b)** ice
(c) clouds **(d)** snow

7. The ozone layer protects us from: 1

(a) infrared radiation **(b)** sound
(c) harmful ultraviolet radiation **(d)** visible light

8. A biogeochemical cycle is the: 1

(a) cycle of seasons
(b) cycle of day and night
(c) cycling of nutrients through living and non-living parts of the Earth
(d) cycle of the Moon

9. All water on the Earth, in oceans, rivers, ice and groundwater, forms the: 1

(a) atmosphere **(b)** biosphere **(c)** hydrosphere **(d)** geosphere

10. The polar regions are colder than the equator mainly because: 1

(a) the Sun does not shine
(b) they have no atmosphere
(c) sunlight falls obliquely and is spread over a larger area
(d) they are nearer to the Sun

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Name the four spheres of the Earth system and state one feature of each. 2

12. What is a biogeochemical cycle? 2

13. State two roles of the atmosphere in supporting life. 2

14. What is solar radiation? Why is it important? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the four spheres interact with an example of a rainy day. 3

16. Explain why the Earth's temperature at the equator and poles differ with the angle of sunlight. 3

17. Explain the importance of the nitrogen-fixing bacteria in agriculture. 3

OR

Describe the natural greenhouse effect and the enhanced greenhouse effect. 3

18. Explain the role of albedo in the Earth's climate with examples of surfaces. 3

OR

Explain how forests help to reduce global warming. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the Earth as a system, describing the four spheres, the flow of energy and the cycling of matter. 5

OR

Describe the nitrogen cycle with the processes: fixation, absorption, assimilation, decay, and denitrification. 5

20. Describe the solar radiation reaching the Earth: speed, time, distribution with latitude and the fate of the radiation (reflected, absorbed, re-radiated). 5

OR

Explain the albedo and the ice-albedo feedback, with the effect of land use change (forests to farmland, urban surfaces). 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND